

READING / RESEARCH NOTE

THE GOLD DOSSIER v1 — 2026-08-24 (run-analyst-golddossier1)

Author: Dr. Mike Darwin (analyst). **Requested by:** the CEO ("Can we try out Gold as an asset?"), third analytical-muscle pick after META and ETH. ****Filed verbatim by the chair at resolve 2026-08-24.** Chair spot-checks against the cached raw data before filing: the expense decay (−0.3964%/yr)

and the 2021+ premium statistics (n=1413, mean −0.0074%, mean|abs| 0.0235%) reproduce to four decimals. Full reproduction scripts in the session scratchpad gold/; run record run-analyst-golddossier1.**

TL;DR

Gold is cheap to own, deeply liquid, and genuinely different from what we already hold – but I found no tradeable edge in it: all four flow stories (central banks, ETF creations, the futures roll, COT positioning) are either backward-looking or statistically dead on 21 years of data.

The one thing worth acting on is a risk parameter, not a trade: 58% of gold's daily risk happens overnight while our venue is shut, so a stop-loss can only reach 42% of it – and gold does NOT protect the book on equity crash days (it averaged −0.26% on the S&P's twenty worst sessions).

Gold's entire 11-year advantage over stocks comes from the last 4.6 years, and my own macro model says 90 percentage points of that move is unexplained by rates or the dollar – that is either a new regime or the thing that unwinds.

Context the CEO should have before anything: gold set an all-time high on 29 Jan 2026, fell 10.3% the next day (its worst session in 22 years), bottomed 26.4% down in July, and is still 14.6% below the high.

Nothing here needs a human decision. This feeds Stan's risk parameters and Ed's menu. No trade is proposed.

Grading key: TRUE-AND-TRADEABLE · TRUE-AND-NOT-TRADEABLE · FALSE · CANNOT TELL. **Fence:** the 2026-08-17 GLD phantom-fill cohort is contaminated and is cited in this dossier for ONE fact only — that GLD fills fractionally through our connector. No P&L, no cost, no comparison from that cohort appears anywhere below. **Host discipline:** all work serial, zero containers, zero bulk extraction; free RAM 1.08–1.20 GB of 15.16 throughout.

o. CONTRARY FACTS FIRST (Darwin's rule)

C1. Gold's entire recent superiority is one 4.6-year window, and my own model cannot explain it. Measured on our feed (2015-08-03 → 2026-08-21, adjusted closes, excess of daily Fed Funds):

	2015-08-03 → 2021-12-31 (6.41Y)	2022-01-03 → 2026-08-21 (4.61Y)
GLD CAGR / vol / Sharpe(excess)	8.04% / 13.89% / 0.560	22.14% / 19.08% / 0.937
SPY	15.75% / 18.10% / 0.849	12.25% / 17.47% / 0.522
DBC	5.18% / 17.05% / 0.328	12.51% / 19.29% / 0.502
TLT	5.14% / 13.82% / 0.365	−8.25% / 15.76% / −0.720

Before 2022 gold lost to equities on risk-adjusted excess return. Anyone quoting the full 11-year Sharpe (GLD 0.729 vs SPY 0.717) is quoting a window that contains the melt-up.

C2. Gold is NOT an equity-crash hedge on our data. On SPY's 20 worst sessions in 11 years (SPY mean −5.20%), **GLD averaged −0.26% and was up on only 9 of 20.** TLT averaged +0.63%. Gold diversifies by LOW correlation (+0.073), not by negative correlation — and the rolling 250-day corr(GLD, SPY) has drifted from **−0.295 (2016) to +0.146 (2026).**

C3. Central banks are price-SENSITIVE, and the World Gold Council says so in its own words. WGC, Gold Demand Trends Q2 2026, published 30 July 2026, verbatim: *"The wider geopolitical backdrop, as well as softer gold prices, are likely to have provided some support for the increased Q2 buying."* They bought MORE because it got cheaper. That is a dip-buyer, not a price-taker.

- C4. Central-bank buying did not stop a 26% drawdown.** Central banks bought a record 289t in Q2 2026 — the same quarter that contained gold's trough. GLD peaked at \$495.90 on **2026-01-29**, fell **–10.27% the next session** (its worst day in 22 years), and bottomed at \$364.96 on **2026-07-16**, **–26.40%** from the high. It is **–14.63%** below the high as of 2026-08-21.
- C5. Gold ETF holdings are still below their 2012 peak.** GLD held 1,353.3 tonnes at end-2012 and holds **1,047.2 tonnes today**, with gold roughly 3.4× higher. Whatever has driven this bull market, it is not Western ETF investors piling in.
- C6. The one number that ends most gold arguments: real rates explain 4.1% of gold's daily variance.** Not 40%. Four.

1. THE INSTRUMENT MAP

1.1 IDENTITY CHECK BEFORE HISTORY (CLAUSE 4), AND IT HOLDS

	OUR FEED'S FIRST BAR	ISSUER INCEPTION (PRIMARY)	NAV/SHARE (ISSUER, 2026-08-21)	OUR LAST CLOSE
GLD	2004-11-18	"Nov 18 2004" (SSGA)	\$420.28	\$423.36
GLDM	2018-06-26	"Jun 25 2018" (SSGA)	\$90.64	\$91.32

Sources: SSGA GLD and GLDM product pages. Independent cross-check: the trust's own historical archive gives Total Net Asset Value **\$154,240,992,702.79** on 2026-08-21; SSGA's page states **"\$154,240.99 M"**. Exact agreement across two independent surfaces. The archive's first row (2004-11-18) has oz/share exactly 0.10000000 and NAV/share \$44.20, implying spot **\$442.00/oz** — the correct gold price at GLD's launch.

1.2 LIQUIDITY AND COST — MEASURED, NOT QUOTED

Median daily dollar volume from our own feed (close × volume):

ETF	\$ADV, LAST 252 SESSIONS	\$ADV, LAST 60	OUR FEED N	FIRST BAR
GLD	\$4,173,840,537	\$2,902,794,094	2,780	2015-08-03
IAU	\$660,234,863	\$416,595,829	2,780	2015-08-03
GLDM	\$445,154,738	\$309,702,239	2,050	2018-06-26
SGOL	\$170,791,164	\$83,543,683	2,780	2015-08-03
IAUM	\$118,516,860	\$79,014,281	1,291	2021-07-01
AAAU	\$78,638,755	\$53,069,322	2,015	2018-08-15
OUNZ	\$34,400,863	\$23,623,149	2,780	2015-08-03
BAR	\$19,958,025	\$7,430,595	2,254	2017-09-01
(for scale) DBC	\$17,516,985	\$23,458,257		
(for scale) DBA	\$9,934,136	\$21,733,008		

The thinnest gold ETF here trades more per day than either commodity ETF we already own. Capacity is not a constraint at any conceivable size.

Expense ratios, MEASURED from realised return differentials:

ETF	PUBLISHED ER	MEASURED VS GLD, 2018-06-26+ (8.13Y)	MEASURED VS GLD, 2021-07-01+ (5.12Y)	IMPLIED
GLD	0.40% (SSGA verbatim)	baseline	baseline	—
IAU	0.25% (secondary)	+18.3 bps/yr	+18.1 bps/yr	~0.22%
GLDM	0.10% (SSGA verbatim)	+31.0 bps/yr	+35.0 bps/yr	~0.07%
SGOL	not verified	+25.5 bps/yr	+27.0 bps/yr	~0.14%
BAR	not verified	+26.0 bps/yr	+26.3 bps/yr	~0.14%

AAAU	not verified	—	+26.4 bps/yr	~0.14%
OUNZ	not verified	+13.4 bps/yr	+16.8 bps/yr	~0.24%
IAUM	not verified	—	+39.3 bps/yr	~0.02%

And a third, cleaner measurement of GLD's fee that uses no price at all. The trust publishes ounces of gold per share daily. It has decayed from **0.10000000 (2004-11-18) to 0.09172109 (2026-08-21)** — 21.75 years — a compound **−0.3964%/yr**, stable across sub-windows (last 10y: −0.3959%/yr; last 5y: −0.3934%/yr). The prospectus fee is 0.40%. **The fund's own physical accounting confirms the fee to within 4 basis points.** (*Chair spot-check at resolve: reproduces exactly from the cached archive.*)

Grade: TRUE-AND-TRADEABLE. GLDM is the cheapest deployable gold wrapper by ~31 bps/yr against GLD, verified two ways.

1.3 WHAT OUR VENUE CAN ACTUALLY TRADE — VERIFIED, WITH THE GAP NAMED

- `alpaca.py:156-161` prices via `StockLatestTradeRequest` — US equities only. Gold ETFs are US-listed equities; the ETH deployability problem does **not** apply.
- `universe.py:118-120` filters on `a.tradable` and `a.fractionable`.
- **GLD is proven tradeable and fractionable at our venue by our own record** (a filled fractional order, qty 0.424471 — cited from the fenced cohort for this single fact only).
- **IAU, GLDM, SGOL tradability at our venue is UNVERIFIED** — needs a live `get_all_assets` call with credentials, a chair action. Absence reported, not assumed. If GLDM is not fractionable, the 31 bps/yr saving is unreachable and GLD is the deployable default.
- Gold ETFs are absent from the hunting ground (a fact about its construction, not about tradability).

1.4 FUTURES (GC) VS ETF — THE BASIS, AND THE SESSION TRAP

The ETH-class alignment defect was checked for and NOT found. GLD daily return vs COMEX front-month (GC=F) at lags −2...+2: $k=0$ corr **+0.9040** (beta +0.8794), all other lags $\leq |0.08|$. $n=2,776$. Metadata confirms identity (FUTURE, COMEX, `firstTradeDate` 2000-08-30).

But the beta is 0.88, not 1.00, and the reason is the most important structural fact in this dossier. Decomposing GLD's day against the same futures day: overnight gap corr **+0.798** (beta 0.590); intraday corr +0.491 (beta 0.291). **Two-thirds of GLD's response to the gold market arrives before our venue opens.** The London fix that sets the ETF's NAV is struck at 10:30 a.m. NYT; gold's price is made in London and Asia while the NYSE is shut.

The basis itself is nothing. Using the LBMA fix recovered from the trust's own data (NAV/share ÷ oz-per-share — the LBMA column was removed at ICE's request but is recoverable by division), COMEX close vs same-date fix, $n=5,461$: mean **+0.014%**, p_5 −0.880%, p_{95} +0.921% (two different clocks — most of the dispersion is 6.5 hours of drift). Largest dislocations: 2026-01-30 (−5.38%), 2008-10-10 (−5.01%), 2008-09-17 (+4.13%), 2006-06-13 (−4.09%), 2008-12-26 (+4.02%), 2020-04-13 (+3.82%) — genuine liquidity events only.

Grade: TRUE-AND-NOT-TRADEABLE. No ETF-vs-futures edge for us; GC=F is the research proxy (2000+), a physical ETF is the deployable instrument, and our venue could not trade GC anyway.

1.5 THE PREMIUM/DISCOUNT, AND A MISTAKE MADE AND CAUGHT

First pass computed GLD's close (\$423.36) against SSGA's NAV (\$420.28) → **+0.73% premium. Artifact.** GLD's NAV is struck at the 10:30 a.m. fix; the close is 16:00 ET. The trust's own like-clock comparison (mid at 16:15 vs indicative at 16:15) reads **0.0057%** the same day.

GLD PREMIUM/DISCOUNT @16:15 ET	N	MEAN	MEDIAN	P5	P95	MEAN ABS
full history	5,469	+0.0030%	—	—	—	—
2021-01-01 onward	1,413	−0.0074%	−0.0052%	−0.0578%	+0.0380%	0.0235%

GLD trades within about two basis points of fair value. (*Chair spot-check at resolve: the 2021+ row reproduces exactly.*)

Method lesson 13: never compute an ETF premium against a NAV struck on a different clock. The magnitude here was **128× the real number**.

Not measured: the bid/ask spread width. The archive gives the mid, not the width; our bars carry no quotes. Absent, reported absent.

2. THE DRIVER MEASUREMENT — gold vs real rates

2.1 PRE-REGISTERED FORM

Beta of gold's daily log return on Δ DFII10 is negative and stable; the claimed 2022–2024 decoupling to be verified, not assumed.

2.2 THE DAILY RELATIONSHIP — REAL, ROBUST, AND SMALL

SERIES	WINDOW	BETA (% PER +1PP REAL YIELD)	T_NW	CORR	R ²	N
GC=F	2003-01-03 → 2026-08-20	−4.535%	−9.99	−0.203	4.1%	5,892
GLD	2004-11-19 → 2026-08-20	−5.612%	−12.75	−0.250	6.2%	5,432

The sign is negative in all 24 calendar years (sign-test $p \approx 6 \times 10^{-8}$). The magnitude is the story: $R^2 = 4.1\%$. Real-rate news explains one twenty-fifth of gold's daily variance.

2.3 THE DOLLAR DOMINATES, AND INFLATION EXPECTATIONS DO NOTHING

MODEL	R ²	D_REAL (%/PP)	D_LNUSD	D_BE (%/PP)
Δ real only	4.10%	−4.535 (t −15.87)	—	—
+ Δ dollar	17.60%	−2.506 (t −9.19)	−90.63 (t −31.06)	—
+ Δ breakeven	17.60%	−2.514 (t −9.12)	−90.67 (t −31.02)	−0.091 (t −0.22)

1. **The dollar is gold's dominant daily driver** (confirmed on our own feed: $\text{corr}(\text{GLD}, \text{UUP}) = -0.4375$, $\text{beta} -1.011$).
2. **"Gold is an inflation hedge" is FALSE at daily frequency** — the breakeven coefficient adds 0.00pp of R^2 . (Scope: tests news, not decade-scale purchasing power.)

2.4 THE 2022–2024 DECOUPLING — VERIFIED, AND IT IS NOT WHAT THE PHRASE IMPLIES

(a) **RETURN correlation did NOT decouple** — 2022 −0.400, 2023 −0.493 (its most negative ever); only 35 of 5,643 rolling windows were ever positive (2009-10). (b) **LEVEL correlation COLLAPSED on schedule** — 2020 −0.958 → 2022-25 −0.106/−0.164/−0.077/−0.059.

Gold kept reacting to real-rate news exactly as always — more strongly than ever — while acquiring a large drift orthogonal to rates. The footprint of a persistent price-taking buyer, not of "gold stopped caring about rates."

Grade: TRUE-AND-NOT-TRADEABLE. The level correlation tells you the regime after it has happened.

2.5 THE SIZE OF THE UNEXPLAINED MOVE — AND THE HONEST POWER STATEMENT

Fit 2003–2021 ($n=4,736$, $R^2=17.83\%$), apply out of sample 2022-01-03 → 2026-08-20 ($n=1,156$):

	CUMULATIVE	ANNUALISED
ACTUAL gold	+147.1%	+21.80%/yr
model PREDICTED	+29.8%	+5.84%/yr
RESIDUAL — unexplained	+90.5%	+15.08%/yr

By year: 2022 +5.02% · 2023 +2.92% · **2024 +26.85%** · **2025 +37.77%** · 2026 +0.83%. **Placebo clean:** fit 2003–16, test 2017–21 → −0.78%/yr.

THE MDE (clause 5). Residual sd 1.110%/d, n=1,156 → |t|=2 needs 16.5%/yr; measured 15.08%/yr → **t = +1.71. The largest macro dislocation in modern gold history does not clear a two-sigma bar.** Confirming it needs ~5.5 years of daily data. **Any "gold has entered a new regime" claim is structurally unconfirmable inside the horizon that matters.** If the residual is permanent, current levels are the baseline; if it fully unwinds, gold falls **−47.5%** from here. Both tails live; no available test discriminates.

2.6 POINT-IN-TIME HONESTY — WHAT THE FRED KEY BOUGHT, MEASURED

Vintage-vs-today comparisons with the new key:

SERIES	AS-OF VINTAGE	OBSERVATIONS	CHANGED	MAX ABS DELTA
DFII10	2022-06-30	4,878	0 (0.00%)	0.0000
DFII10	2024-12-31	5,503	0 (0.00%)	0.0000
DTWEXBGS	2022-06-30	4,134	1,833 (44.34%)	0.2060
DTWEXBGS	2024-12-31	4,761	1,723 (36.19%)	0.4167
VIXCLS	2024-12-31	5,557	0 (0.00%)	0.0000

1. **The real-rate centerpiece is point-in-time clean** — DFII10 is never revised. The keyless-feed warning is real for macro AGGREGATES and not for market-price series; that distinction was a guess before today and is a measurement now.
2. **The dollar index IS revised** (36–44% of observations) — the dollar leg of \$2.3 carries revision contamination the real-rate leg does not. Stated, not papered over.
3. **The genuine PIT constraint is a RELEASE lag, not a revision:** the 2022-06-30 vintage of DFII10 lacks 2022-06-30 itself — **day D's real yield is not visible on day D.** Any same-day rule is taking a few hours of look-ahead.
4. Current lags (2026-08-24): DFII10 → 08-20 · DTWEXBGS → 08-14 · CPIAUCSL → 07-01 · FEDFUNDS → 07-01.

3. NAMED COUNTERPARTIES — four flow stories, each priced

3.1 CENTRAL BANKS — REAL, LARGE, PRICE-SENSITIVE, STRUCTURALLY UNMEASURABLE FOR US

WGC Q2 2026 (published 30 July, data to 24 July, revises): net purchases **289t**; H1 345t — *"the lowest for a first half since 2022."* Poland +51t (to 632t), China +33t, Uzbekistan +16t, Kazakhstan +15t. Secondary press reported the YoY jump as "74%"; the primary's figure is **62%** (289 ÷ 177.9 = +62.4%). **Use the primary.**

Tradeable behind a 30-day lag? No — and the reason is POWER, computed before any test: quarterly return sd ≈ 8.1%, ~100 usable quarters, tercile MDE **4.0%/quarter = 17%/yr.** No plausible flow signal delivers that. The test cannot mean anything, so it was not run. **The falsifier is already half-fired:** H1 2026 was the weakest since 2022 — the flow is not monotonic.

Grade: TRUE-AND-NOT-TRADEABLE.

3.2 ETF CREATIONS/REDEMPTIONS — THE BEST DATASET IN THIS DOSSIER, AND A CLEAN KILL

Source (a trap on the way: the advertised .csv **301-redirects to a PDF** — check Content-Type, not extension): api.spdrgoldshares.com/api/v1/historical-archive?product=gld&exchange=NYSE&lang=en (XLSX, browser UA required) — **5,472 usable rows, 2004-11-18 → 2026-08-21** (204 "US Holiday" string rows dropped): oz/share, NAV@10:30, indicative@16:15, mid@16:15, premium, volume, ounces, tonnes, total NAV.

Publication lag from the sponsor's own page: day D's holdings post after the 16:00 close → earliest tradeable action is D+1's open, which is exactly what the test uses. Flow structure: 47.6% of days zero change; median |flow| 0.30t; max 49.76t; tonnes 8.1 (2004) → 1,353.3 (2012) → **1,047.2 today.**

HORIZON	N	**PRE-STATED MDE AT \	T\	=2**	HIGH-MINUS-LOW FLOW TERCILE	T
next session	5,219	0.079%	−0.038%	−0.96		
1 week	5,215	0.173%	−0.118%	−1.38		

1 month	5,199	0.344%	-0.232%	-1.36
contemporaneous (not tradeable)	5,220	—	+0.404%	+10.19

ETF flow follows the price. It does not lead it. Replicates the ETH ETF-flow kill on a 21-year sample. Only a genuinely predictive creation signal (intraday AP orders, AP inventory — neither public) would change this.

Grade: FALSE as a signal, TRUE-AND-TRADEABLE as an instrument-quality dataset.

3.3 THE FUTURES ROLL — NOBODY PAYS IT, AND THAT IS THE FINDING

Gold is a pure carry market: contango $\approx$ financing. Net roll cost to a fully-collateralised long $\approx$ **zero in every rate regime** (2021: 0.01%/roll; 2023: 0.84% recovered on collateral; 2026: 0.61% recovered). **The true cost of the ETF wrapper is the fee PLUS ~3.6%/yr of forgone collateral income at today's funds rate** — nine times the expense ratio. That makes the ETF a *fully-funded* position (what an unlevered mandate wants), and makes any ETF-vs-futures return comparison that ignores it a funded-vs-levered comparison.

Grade: TRUE-AND-NOT-TRADEABLE.

3.4 COT POSITIONING — FREE, MACHINE-READABLE, AND DEAD

CFTC Socrata, code 088691, **1,930 reports 1986-01-15 → 2026-08-18** (Tuesdays 1,768). Release Fridays 15:30 ET; entry rule the following Monday (no look-ahead). Latest: OI 406,260 contracts (~\$190.6bn), non-commercial net +222,189.

HORIZON	N	PRE-STATED MDE	LOW-MINUS-HIGH TERCILE	T
1 week	1,234	0.35%	+0.012%	+0.07
1 month	1,230	0.69%	-0.185%	-0.54
3 months	1,221	1.13%	-1.494%	-2.56 ←

The 3-month result is an overlap artifact, killed properly: 13 disjoint non-overlapping phases give -1.88 -2.35 -1.46 -0.89 -0.10 +0.27 +0.63 +0.17 +0.03 -0.19 -1.38 -1.09 -1.40 — **1 of 13 reaches |t|>2, signs disagree**, and the pooled t was inflated $\sqrt{12.6} \approx 3.55\times$ by overlap. What little signal exists is *momentum*, the opposite of the "fade the specs" story.

Grade: FALSE. Revivable only by the Disaggregated managed-money subseries (2006+) or a non-public intra-week proxy.

4. RISK PARAMETERS FOR STAN

(Per the accepted routing: risk parameters and menu inputs; no position, no candidate.)

4.1 THE HEADLINE PARAMETER: 58% OF GOLD'S DAILY RISK IS OUTSIDE OUR REACH

Variance decomposition, 2015-08-03 → 2026-08-21, n=2,779 (OHLC share one adjustment factor, marketdata.py:266-273, so the split is valid):

INSTRUMENT	ANN VOL	OVERNIGHT GAP SD/DAY	INTRADAY SD/DAY	GAP SHARE
GLD	16.27%	0.778%	0.625%	57.6%
IAU	16.24%	0.777%	0.625%	57.7%
GLDM	17.21%	0.824%	0.654%	57.7%
SLV	31.60%	1.537%	1.197%	59.6%
SPY	17.84%	0.729%	0.847%	42.1%
TLT	14.68%	0.683%	0.638%	54.6%
DBC	18.01%	0.824%	0.779%	52.8%
DBA	13.00%	0.499%	0.643%	37.2%
GDX	39.19%	1.505%	2.005%	37.2%

Bootstrap 95% CI on GLD's gap share: **[53.4%, 62.0%]**. **A stop-loss on gold reaches at most 42% of its risk.** Gap distribution: mean +0.0557%, sd 0.778%, p1 -2.13%, p99 +2.07%, min **-5.98%** (2026-01-30). Monday-gap ratio only 1.067 — no weekend, unlike crypto. GDX inverts the profile (37% gap) because it IS a US equity — stop-reachable gold exposure exists only at 39% vol and -49.8% drawdowns.

4.2 VOLATILITY REGIME — GOLD IS AT TWICE ITS NORMAL VOLATILITY RIGHT NOW

2017 9.90% · 2019 11.66% · 2021 13.68% · 2023 13.39% · 2024 15.01% · 2025 **19.85%** (+63.68% return) · **2026 to 08-21: 32.14%** (+6.83%). **Any vol parameter inherited from a pre-2025 study is wrong by a factor of two today.**

4.3 DRAWDOWN GEOMETRY

ATH close \$495.90 (2026-01-29) → **-10.27% next session** (worst in 22y) → trough \$364.96 (2026-07-16, **-26.40%**) → now -14.63%. Worst sessions ever: 2026-01-30 -10.27% · 2013-04-15 -8.78% · 2008-10-10 -7.52% · 2006-06-13 -6.85% · 2025-10-21 -6.43%. Max DD (11y feed): GLD -26.40% vs SPY -33.72% / DBC -41.71% / TLT -48.35% / GDX -49.79% / SLV -52.28%. The 2026-01-30 *cause* (Warsh nomination after a parabolic January) is secondary-sourced: **CANNOT TELL on causation; TRUE on the price action.**

4.4 CORRELATION TO THE EXISTING BOOK, AND THE DBC OVERLAP MEASURED HONESTLY

corr(GLD, ·), n=2,779: SPY **+0.073** · TLT +0.258 · DBC **+0.256** · DBA +0.122 · TIP +0.347 · **UUP -0.4375 (beta -1.011)** · GDX +0.774 · SLV +0.780.

The brief's "DBC carries ~8-12% gold" premise could NOT be verified at a primary — reported **ABSENT** (Invesco confirms gold is one of 14 components; weights unobtainable). The overlap itself is measured and small: DBC on GLD → R² **6.54%**. Adding gold beside DBC is not doubling an exposure.

Crisis behaviour:

SPY'S WORST K SESSIONS	SPY MEAN	GLD MEAN	TLT MEAN	DBC MEAN	GLD UP-DAYS
20	-5.20%	-0.26%	+0.63%	-2.59%	9 / 20
50	-3.91%	-0.08%	+0.62%	-1.70%	25 / 50
100	-3.08%	+0.09%	+0.49%	-1.14%	47 / 100

Gold is a coin flip on equity crash days. Treasuries are the thing that rises.

4.5 BOOK IMPACT, AT THE LIVE NAV (\$1,885.74, CASH 51.4%)

CONFIGURATION	ANN VOL	DELTA
current book	4.52%	—
+ GLD 5% NAV from cash	4.84%	+0.32pp
+ GLD 5% from DBC	4.23%	-0.29pp
+ GLD 10% from cash	5.27%	+0.74pp
+ GLD 10% from DBC	4.19%	-0.33pp
+ GLD 15% from cash	5.78%	+1.26pp

The funding source matters more than the size. Worst-case arithmetic at caps: 5% NAV → worst session -\$9.69 (-0.51% NAV), worst gap -\$5.64; 10% → -\$19.37 / -\$11.28; 15% → -\$29.06 / -\$16.92.

4.6 MDE CONSTANTS FOR ANY FUTURE GOLD EVENT WORK (REUSE; DO NOT RE-DERIVE)

Raw daily sd **1.025%**. Residual sd: vs SPY **1.022%** (SPY removes 0.3% of the variance — a useless benchmark), vs DBC 0.991%, vs GDX 0.649%, vs macro model 1.11-1.13%.

N EVENTS	N=5D MDE	N=20D MDE
10	1.45%	2.90%

50	0.65%	1.30%
250	0.29%	0.58%

Most per-event gold hypotheses (FOMC/CPI/payrolls on one instrument) are structurally unmeasurable. Route to the cross-section or the bin.

5. PRE-REGISTERED PRIOR: volatility targeting on gold

Registered 2026-08-24, before any volscale-style gold candidate exists. Harvey et al. 2018: vol targeting improves Sharpe only for risk assets; *"for bonds, currencies, and commodities the impact on the Sharpe ratio is negligible"*, while tail severity improves across all classes.

1. **Sharpe: no improvement expected** — gold shows the *opposite* of the equity leverage effect (its highest-vol years were its best: 2025 19.85% vol / +63.68%). A vol-targeted sleeve systematically underweights gold in its best years. Expect ≤ 0 .
2. **Tail: genuine improvement expected** — gold's worst sessions cluster in high-vol regimes.
3. **THE BAR, computed first:** $SE(\text{Sharpe}) \approx 0.357$ on 11y (0.221 on 23y) → **a Sharpe difference below ~0.6 is undetectable on every gold sample we can construct.** "Improved 0.73 → 0.91" is +0.18 against SE 0.36 — noise. The correct claim for such a candidate is the tail claim.

Scoring: P10 in the ledger.

6. DATA-SOURCE AUDIT

Embargo honoured: **no collector output consumed**; every series fetched at a primary or vendor source this dispatch. 12 sources, graded (full table in the run record): our Yahoo-sourced ETF bars (GOOD, adjustment verified), GC=F/DXY chart API (ADEQUATE — front-month splice, daily returns only), FRED keyless (DFI10/VIXCLS EXCELLENT-never-revised; DTWEXBGS CONTAMINATED), FRED vintage with key (GOOD; key never printed), **the SPDR archive (EXCELLENT — issuer primary, 5,472 rows)**, CFTC Socrata (GOOD), WGC Q2 (GOOD as text), SSGA pages (EXCELLENT), iShares (WEAK — 403s automated fetch; the measured differential is the stronger evidence), Invesco DBC weights (**ABSENT**), the fund's own endpoints (GOOD).

Traps recorded: the SPDR .csv URL 301-redirects to a PDF (Content-Type, not extension); FRED's LBMA gold series are discontinued 404s and Yahoo XAUUSD=X 404s — **there is no free FRED gold price today; the recovered NAV÷oz series is the substitute and it is better**; iShares and SSRN 403 automated fetches; 204 archive rows read "US Holiday" in every column; openpyxl is not in the venv.

7. PREDICTION LEDGER — P1 to P10

#	PREDICTION	RESOLVES	SCORES WRONG IF
P1	GLD oz/share on 2026-12-31 = 0.09159 ± 0.00008	2026-12-31	outside band
P2	median 16:15 premium 2026-09-01→12-31 within ±0.02%	2027-01-05	outside
P3	flow-tercile next-session spread \	t\	< 2 on Q4 data 2027-01-05 \ t\ ≥ 2
P4	COT 3-month: fewer than 3 of 13 disjoint phases reach \	t\	≥ 2 through 2027-06-30 2027-07-05 ≥ 3 do
P5	2026 full-year real-rate beta negative (24 of 24 years)	2027-01-05	positive
P6	rolling 250d LEVEL corr stays above −0.60 through 2026-12-31 (decoupling persists)	2027-01-05	any print ≤ −0.60
P7	WGC Q3 central-bank net purchases in [+100t, +400t], published by 2026-11-07	2026-11-07	outside, or late

P8	GLD median \$ADV Sep–Dec > \$1.5bn	2027-01-05	≤
P9	GLDM beats GLD by +25 to +40 bps over 12mo to 2027-08-21	2027-08-23	outside
P10	any belt gold vol-target candidate: Sharpe delta < +0.6 AND worst-20d drawdown improves > 15%	2027-06-30 or void	either fails

8. THE HONEST VERDICT

As an INSTRUMENT: TRUE-AND-TRADEABLE, no reservations. Fractionally fillable at our venue (proven by our own record), \$4.17bn/day ADV, fee verified three independent ways (0.40% GLD / ~0.10% GLDM), ~2 bps from fair value, no basis to fight, no roll to pay, no unlock calendar, no counterparty. Cleaner, cheaper, more liquid than anything the fund currently holds.

As an EDGE: nothing found, on 21 years of free primary data. Four named counterparties, four honest kills, zero manufactured signals.

What survives — risk parameters, not trades: (1) 58% of daily variance is overnight [53.4%, 62.0%] — size off the gap distribution; (2) gold is not an equity-crash hedge (−0.26% on SPY's worst 20; corr drifted −0.295 → +0.146); (3) funding dominates sizing (10% from cash: book vol 4.52→5.27%; from DBC: 4.19%); (4) 2026 realised vol 32.14% — double the decade average — 14.6% below an ATH set seven months ago after a −26.4% drawdown.

And the contrary fact not to lose: +90.5pp of the last 4.6 years is unexplained by rates, the dollar, or inflation expectations — at t = +1.71, unconfirmable either way for ~5.5 more years. Anyone sizing gold today is sizing an unconfirmed regime break, and the honest place to say so is in the sizing, not the thesis.

The money question, answered directly: at \$1,885 NAV, a 10% sleeve is \$188.57; its worst historical session costs \$19.37, its worst gap \$11.28. Affordable — but **this dossier produces no alpha claim and therefore no candidate.** Gold here is a **beta allocation decision** — Stan's judgement on the parameters above, the CEO's click. *True, and tradeable only as an allocation.*

Reproduction: scripts g1–g19 + cached data in the session scratchpad goLd/. Chair's resolve actions and the seat's STATE/BINDS are in run-analyst-golddossier1 and .claude/state/analyst.md. Per the non-negotiables this document is never edited.